Wireless Charging Facility Deployment and Adaptive Battery Configuration under Service-Plan Uncertainty

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Abstract

This abstract will be written after the main body is stable.

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1. Introduction

Electrifying bus fleets requires coordinated planning of charging infrastructure, battery capacity, and charging-resource usage under future service requirements. Wireless charging technologies expand the set of charging opportunities by allowing buses to receive energy at selected stops or road links during regular service. This capability turns infrastructure planning from a stand-alone siting decision into a joint planning problem involving wireless facility deployment, route-level battery configuration, and the use of existing plug-in fast chargers.

Wireless charging facilities are long-lived assets, whereas transit service plans may change with future demand, fleet composition, and operating policies. A layout that is attractive for one future service plan may perform poorly under another because buses may operate different trip chains, require different battery capacities, and compete for existing fast-charging resources at different times. The infrastructure decision therefore has to be made before the future fleet-service configuration is known, whereas battery-capacity and charging-resource decisions can adapt after that configuration is realized.

The central planning trade-off is shaped by the interaction among wireless infrastructure, battery capacity, and plug-in fast-charging capacity. More in-route wireless charging can reduce battery requirements on some routes and shift energy replenishment away from terminals or depots. At the same time, wireless deployment changes the timing and amount of residual plug-in fast charging needed by different routes. Because plug-in fast chargers are limited shared resources, these residual demands couple route-level decisions that would otherwise be separable.

We capture this timing and coupling structure with a two-stage stochastic mixed-integer planning model. The first stage selects stationary and dynamic wireless charging facilities. The second stage, after a fleet-service scenario is realized, chooses scenario-adaptive battery configurations and representative-day charging schedules. The resulting recourse problems contain discrete battery choices, charging-scheduling decisions, and shared fast-charger capacity constraints. Consequently, both the deterministic equivalent and the classical integer L-shaped decomposition become difficult to scale as the network and scenario set grow.

To solve this model, we develop a reformulation-enabled integer L-shaped method. The method differs from a standard integer L-shaped decomposition by changing what recourse information is exposed to the master and by separating inexpensive lower-bound generation from exact recourse verification. This design preserves exactness when the cut loop converges and creates the algorithmic components evaluated in the computational study.

The paper makes three contributions. First, it identifies a two-stage stochastic mixed-integer structure in which here-and-now facility decisions interact with scenario-dependent discrete design choices and shared-resource operational recourse. The wireless-charging application provides a concrete instantiation of this structure, and the formulation highlights the coupling that drives the algorithmic development. Second, it develops an exact reformulation-enabled integer L-shaped algorithm that combines scenario-dependent lifting, a two-phase relaxation-verification scheme, and monotonicity-strengthened cuts for mixed-integer recourse. Third, it provides a computational performance analysis that evaluates scalability with respect to network size and scenario count, and tests the method's adaptability when the gap between the Phase 1 relaxation and Phase 2 exact verification varies.

The experiments use wireless-charging planning instances constructed from Singapore bus-service data for routes operated by Tower Transit. They are designed to answer two questions: whether the model produces interpretable planning decisions, and which algorithmic components expand the solvable instance boundary. They also vary the available plug-in fast-charging capacity to test how the proposed method behaves as shared-resource coupling becomes tighter or looser.

The remainder of the paper is organized as follows. Section 2 reviews related work on electric-bus charging infrastructure planning and stochastic decomposition methods. Section 3 presents the two-stage formulation and the baseline integer L-shaped decomposition. Section 4 develops the lifted two-phase algorithm and strengthened cuts. Section 5 reports the computational results. Section 6 concludes.

2. Literature Review

This section reviews two streams of literature that define the paper's application structure and algorithmic positioning. The first stream is electric-bus charging and wireless infrastructure planning, which motivates the interaction among wireless facilities, battery choices, and existing plug-in fast-charging resources. The second stream is stochastic facility-location modeling and decomposition, which provides the methodological context for two-stage mixed-integer planning with scenario-dependent recourse. The review is organized to keep the transportation discussion concise and to focus on the shared-resource mixed-integer recourse structure used by the proposed algorithm.

2.1. Electric-bus charging and wireless infrastructure planning

Electric-bus charging studies commonly coordinate infrastructure deployment, battery sizing, vehicle or charging schedules, and service feasibility. These decisions are often formulated as integrated optimization models because charging resources, route energy requirements, and schedule feasibility interact. Depot and plug-in fast-charging studies emphasize charger availability and charging schedules, whereas opportunity-charging studies add charging possibilities during service.

Wireless charging infrastructure planning extends this setting by considering stationary wireless charging at stops and dynamic wireless charging on road links. This literature provides modeling elements that are central to our application: candidate wireless facilities, route-level energy feasibility, battery-capacity implications, and charging-operation constraints. It also motivates the coexistence of wireless facilities with existing plug-in fast-charging resources.

Our use of this stream is structural. We adopt wireless-facility placement, route-level battery choices, representative-day charging operations, and existing plug-in fast chargers as the application objects of the model. We do not focus on real-time charging control, bus dispatch, or traffic operations; instead, these objects define a long-term planning setting whose shared-resource mixed-integer recourse structure is reviewed methodologically next.

2.2. Stochastic facility-location models and decomposition methods

Stochastic facility-location and network-design models provide the optimization background for here-and-now infrastructure decisions under uncertain future conditions. In these models, scenarios typically represent uncertain demand, service requirements, or operating conditions, and the recourse problem evaluates the consequence of a first-stage design after a scenario is realized. Benders decomposition, integer L-shaped methods, and logic-based Benders decomposition provide the standard decomposition language for such two-stage models, especially when the deterministic equivalent is too large or the recourse problem contains discrete decisions.

Within this decomposition family, computational performance is shaped by two broad design dimensions. The first is the decomposition or reformulation choice: what information is retained in the master problem, what remains in the scenario subproblems, and whether the subproblems can be decomposed or relaxed without losing useful bound information. The second is the separation strategy: how feasibility and optimality information from the subproblems is converted into cuts, and whether those cuts exploit problem-specific structure beyond a basic no-good exclusion.

Research on decomposition for mixed-integer recourse has developed techniques along both dimensions. On the reformulation side, stronger master representations, partial recourse descriptions, relaxed subproblems, and internal subproblem decompositions can improve lower bounds or reduce separation cost. On the cut side, integer L-shaped cuts, logic-based Benders cuts, combinatorial cuts, valid inequalities, and dominance-based cuts can exploit structure that is not captured by generic linear duality. The effectiveness and validity of these enhancements are structure dependent, which is why successful decomposition algorithms are often tailored to the recourse model rather than transferred directly from one application to another.

This paper follows that tailored-decomposition perspective. The electric-bus wireless-charging setting creates a shared-resource mixed-integer recourse structure with scenario-dependent battery choices, route-level operating decisions, and plug-in fast-charger capacity coupling. The proposed method instantiates the two design dimensions above through a reformulation-enabled integer L-shaped algorithm: it changes the master–subproblem organization, uses a relaxation–verification separation scheme, and derives cuts from the joint facility–battery structure while retaining exact recourse verification.

3. Problem Formulation and Recourse Structure

This section formulates the wireless charging infrastructure planning problem as a two-stage stochastic program. The formulation separates long-lived wireless-facility decisions from scenario-adaptive battery and charging decisions, and then identifies the recourse structure used by the algorithm in Section 4.

3.1. Two-stage stochastic formulation

Let Ω be the set of service-plan scenarios, and let p_ω be the probability of scenario $\omega \in \Omega$, where $p_\omega \geq 0$ and $\sum_{\omega\in\Omega} p_\omega = 1$. Each scenario represents one plausible future service plan and fixes the route frequencies, timetables, bus trip chains, active-fleet requirements, and trip-level energy requirements, while the operated route set $\mathcal{R}$ is common across scenarios. The planner first installs long-lived wireless charging facilities before the future service plan is known. After a scenario is realized, the second-stage recourse selects route-level battery capacities and the corresponding representative-day charging allocation and energy-management decisions.

Let $\mathcal{S}$ be the station set, $\mathcal{L}$ the directed link set between consecutive stations, and $C \subseteq \mathcal{S}$ the set of existing interchanges or hubs equipped with plug-in chargers. Stationary wireless charging (SWC) can be installed at any station in $\mathcal{S} \setminus C$, while dynamic wireless charging (DWC) can be installed on any directed link in $\mathcal{L}$. The first-stage binary variables w_s^{S} and w_l^{D} indicate whether SWC is installed at station $s \in \mathcal{S} \setminus C$ and DWC is installed on link $l \in \mathcal{L}$, respectively. Existing plug-in chargers at stations in C are pre-existing resources with limited shared capacity and are not candidates for SWC deployment.

The compact two-stage stochastic formulation is

$$\min_{w^{\mathrm{S}},w^{\mathrm{D}}} \quad C^{\mathrm{fac}}(w) + \sum_{\omega\in\Omega} p_\omega \Phi_\omega(w) \tag{1}$$

$$\text{s.t.} \quad w_s^{\mathrm{S}} \in \{0,1\}, \qquad s \in \mathcal{S} \setminus C, \tag{2}$$

$$w_l^{\mathrm{D}} \in \{0,1\}, \qquad l \in \mathcal{L}. \tag{3}$$

where $w = (w^{\mathrm{S}}, w^{\mathrm{D}})$ and

$$C^{\mathrm{fac}}(w) = \sum_{s\in\mathcal{S}\setminus C} c_s^{\mathrm{S}} w_s^{\mathrm{S}} + \sum_{l\in\mathcal{L}} c_l^{\mathrm{D}} w_l^{\mathrm{D}}.$$

Here c_s^{S} and c_l^{D} are the representative-day equivalent costs of installing SWC at station s and DWC on link l, respectively. Facility and battery investment costs are converted to representative-day equivalents to match the charging-cost scale. The vector w is the nonanticipative here-and-now facility decision, and $\Phi_\omega(w)$ is the optimal second-stage value in scenario ω.

The scenario decisions consist of route-level battery selection variables $z_{\omega,r,k}$ and operating variables collected in y_ω, including charging amounts, plug-in occupancy, and battery energy states. For a fixed facility layout w, the scenario recourse value $\Phi_\omega(w)$ is the minimum scenario cost

$$\Phi_\omega(w) = \min_{z_\omega, y_\omega} \left\{ C_\omega^{\mathrm{bat}}(z_\omega) + C_\omega^{\mathrm{chg}}(y_\omega) : (z_\omega, y_\omega) \in \mathcal{X}_\omega(w) \right\}. \tag{4}$$

The next subsection expands $\mathcal{X}_\omega(w)$.

3.2. Scenario recourse formulation

For a fixed scenario ω, the recourse formulation uses route-indexed trip-chain data. For each route $r \in \mathcal{R}$, $\mathcal{M}_{\omega,r}$ is the set of trip chains, $\mathcal{A}_{\omega,r}$ is the set of directed traversals, and $\mathcal{V}_{\omega,r}^{\mathrm{P}}$ and $\mathcal{V}_{\omega,r}^{\mathrm{S}}$ are the plug-in and SWC station visits. Each trip chain corresponds to one bus, so $n_{\omega,r} = |\mathcal{M}_{\omega,r}|$ is the route fleet size in scenario ω. A consolidated notation table is provided in Appendix Appendix A.

Recourse costs.. Battery options $\mathcal{K}$ have capacities β_k, with representative-day equivalent unit cost c^{bat}. Operating cost is based on grid-side energy purchased through plug-in charging, stationary wireless charging, dynamic wireless charging, and overnight depot charging. The corresponding charging amounts are q_v^{P}, q_v^{S}, q_a^{D}, and q_m^{N}, with daytime and overnight electricity prices c^{day} and c^{night}. The scenario battery cost is

$$C_\omega^{\mathrm{bat}}(z_\omega) = c^{\mathrm{bat}} \sum_{r\in\mathcal{R}} n_{\omega,r} \sum_{k\in\mathcal{K}} \beta_k z_{\omega,r,k}.$$

The scenario charging cost is

$$C_\omega^{\mathrm{chg}}(y_\omega) = c^{\mathrm{day}} \left(\sum_{r\in\mathcal{R}} \sum_{v\in\mathcal{V}_{\omega,r}^{\mathrm{P}}} q_v^{\mathrm{P}} + \sum_{r\in\mathcal{R}} \sum_{v\in\mathcal{V}_{\omega,r}^{\mathrm{S}}} q_v^{\mathrm{S}} + \sum_{r\in\mathcal{R}} \sum_{a\in\mathcal{A}_{\omega,r}} q_a^{\mathrm{D}} \right) + c^{\mathrm{night}} \sum_{r\in\mathcal{R}} \sum_{m\in\mathcal{M}_{\omega,r}} q_m^{\mathrm{N}}.$$

Battery configuration.. The route-level battery-choice constraints are

$$\sum_{k\in\mathcal{K}} z_{\omega,r,k} = 1, \qquad r \in \mathcal{R}, \tag{5}$$

$$B_{\omega,r} = \sum_{k\in\mathcal{K}} \beta_k z_{\omega,r,k}, \qquad r \in \mathcal{R}, \tag{6}$$

$$z_{\omega,r,k} \in \{0,1\}, \qquad r \in \mathcal{R},\ k \in \mathcal{K}. \tag{7}$$

Constraints (5)–(7) select one common battery option for all buses assigned to each route and define the induced route battery capacity $B_{\omega,r}$.

Shared plug-in charging.. Plug-in charger use is modeled over scenario time slots $\mathcal{T}_\omega$. A visit v can use plug-in charging only during $\mathcal{T}_v \subseteq \mathcal{T}_\omega$, and $s(v)$ maps the visit to its station. The binary variable $x_{v\tau}$ records whether visit v occupies a plug-in charger in time slot τ; each occupied slot delivers up to κ units of grid-side energy, and station s has capacity U_s. The plug-in charging constraints are

$$0 \le q^{\mathrm{P}}_v \le \kappa \sum_{\tau\in\mathcal{T}_v} x_{v\tau}, \qquad r \in \mathcal{R},\ v \in \mathcal{V}^{\mathrm{P}}_{\omega,r}, \tag{8}$$

$$\sum_{r\in\mathcal{R}} \sum_{\substack{v\in\mathcal{V}^{\mathrm{P}}_{\omega,r}:\\ s(v)=s,\ \tau\in\mathcal{T}_v}} x_{v\tau} \le U_s, \qquad s \in C,\ \tau \in \mathcal{T}_\omega, \tag{9}$$

$$x_{v\tau} \in \{0,1\}, \qquad r \in \mathcal{R},\ v \in \mathcal{V}^{\mathrm{P}}_{\omega,r},\ \tau \in \mathcal{T}_v. \tag{10}$$

Constraints (8)–(10) define plug-in charging amounts, plug-in charger occupancy, and station-time capacity. The shared plug-in charger-capacity constraint (9) provides the main cross-route coupling within a scenario.

Route-local operational feasibility.. Route energy feasibility is written over station visits $\mathcal{V}_{\omega,r}$ and directed traversals $\mathcal{A}_{\omega,r}$. The mappings $\ell(a)$, tail(a), and head(a) connect each traversal to its physical link and adjacent visits. Traversal and depot energy requirements are represented by d_a, d^0_m, and d^N_m, while $\bar{q}$ and η collect charging-opportunity limits and charging efficiencies. The variables e^{arr}_v and e^{dep}_v are the battery energy states at arrival and departure of visit v. The state-of-charge fractions are bounded by $\underline{\rho}$ and $\overline{\rho}$, and v^0_m and v^N_m denote the first and last visits of trip chain m; detailed notation is listed in Appendix Appendix A. Trip-level energy requirements are exogenous and do not vary with the selected battery capacity. Overnight depot charging is assumed to have sufficient power and capacity to restore all buses to the prescribed initial state of charge before the next representative day. Event-level variables suppress scenario and route indices because each visit and traversal is already a scenario-route-specific copy. Charging quantities are nonnegative continuous grid-side energy variables, and battery-energy states are continuous. For each route $r \in \mathcal{R}$, the route-local operational constraints are

$$0 \le q^{\mathrm{S}}_v \le \bar{q}^{\mathrm{S}}_v w^{\mathrm{S}}_{s(v)}, \qquad v \in \mathcal{V}^{\mathrm{S}}_{\omega,r}, \tag{11}$$

$$0 \le q^{\mathrm{D}}_a \le \bar{q}^{\mathrm{D}}_a w^{\mathrm{D}}_{\ell(a)}, \qquad a \in \mathcal{A}_{\omega,r}. \tag{12}$$

$$\underline{\rho} B_{\omega,r} \le e^{\mathrm{arr}}_v \le \overline{\rho} B_{\omega,r}, \qquad v \in \mathcal{V}_{\omega,r}, \tag{13}$$

$$\underline{\rho} B_{\omega,r} \le e^{\mathrm{dep}}_v \le \overline{\rho} B_{\omega,r}, \qquad v \in \mathcal{V}_{\omega,r}, \tag{14}$$

$$e^{\mathrm{dep}}_v = e^{\mathrm{arr}}_v + \eta^{\mathrm{P}} q^{\mathrm{P}}_v, \qquad v \in \mathcal{V}^{\mathrm{P}}_{\omega,r}, \tag{15}$$

$$e^{\mathrm{dep}}_v = e^{\mathrm{arr}}_v + \eta^{\mathrm{S}} q^{\mathrm{S}}_v, \qquad v \in \mathcal{V}^{\mathrm{S}}_{\omega,r}, \tag{16}$$

$$e^{\mathrm{arr}}_{\mathrm{head}(a)} = e^{\mathrm{dep}}_{\mathrm{tail}(a)} - d_a + \eta^{\mathrm{D}} q^{\mathrm{D}}_a, \qquad a \in \mathcal{A}_{\omega,r}, \tag{17}$$

$$e^{\mathrm{arr}}_{v^0_m} = \overline{\rho} B_{\omega,r} - d^0_m, \qquad m \in \mathcal{M}_{\omega,r}, \tag{18}$$

$$e^{\mathrm{dep}}_{v^N_m} - d^N_m \ge \underline{\rho} B_{\omega,r}, \qquad m \in \mathcal{M}_{\omega,r}, \tag{19}$$

$$\eta^{\mathrm{P}} q^{\mathrm{N}}_m = \overline{\rho} B_{\omega,r} - e^{\mathrm{dep}}_{v^N_m} + d^N_m, \qquad m \in \mathcal{M}_{\omega,r}. \tag{20}$$

Constraints (11)–(12) link route operations to first-stage wireless-facility decisions, and constraints (13)–(14) impose arrival and departure state-of-charge bounds. Constraints (15)–(17) update the energy state through station charging and directed traversals, while constraints (18)–(20) close each trip chain through depot movements and overnight recharge.

3.3. Recourse structure

For a fixed facility layout w, the expected recourse term is separable across scenarios. This yields a standard integer L-shaped reference in which each scenario is evaluated as a monolithic mixed-integer recourse problem containing both battery-selection and operational decisions. We refer to this direct decomposition as Natural ILS. The recourse problem also exhibits conditional route locality. Constraints (11)–(20) are imposed independently by route, whereas the shared plug-in charger-capacity constraint (9) couples plug-in occupancy across routes at each station–time pair. Finally, the formulation induces a facility–battery resource order. Under the model specification that trip-level energy consumption is independent of the selected battery capacity, additional wireless facilities and larger route batteries preserve operational feasibility under resource-dominating configurations and cannot increase the minimum charging cost.

These properties define a layered recourse structure comprising scenario-dependent discrete battery configurations, conditionally route-local operations, and limited shared-capacity coupling. Section 4 develops an exact decomposition method tailored to this structure.

4. Reformulation-Enabled Integer L-Shaped Method

This section develops an exact integer L-shaped method that exploits the structure exposed by the model in Section 3. The method first lifts scenario-dependent battery configurations into the master problem, then uses a two-phase separation scheme to combine route-level LP lower bounds with exact mixed-integer verification, and finally strengthens cuts through the monotone facility–battery resource order.

Throughout this section, $Q_\omega(w, z_\omega)$ denotes the exact operational recourse value after the facility layout and scenario battery configuration are fixed, and θ_ω denotes its master approximation. When the distinction between station-based SWC and link-based DWC is not needed, $\mathcal{J}$ denotes the combined index set of first-stage wireless decisions.

4.1. Lifted master reformulation

The baseline integer L-shaped decomposition keeps only the facility vector w in the master and evaluates each scenario through $\Phi_\omega(w)$, which jointly optimizes battery configuration and operations. This representation follows directly from the two-stage information structure, but it hides the battery configuration inside the mixed-integer recourse problem. We instead lift the scenario-dependent battery variables into the master. The variables z_ω remain scenario-adaptive recourse decisions; no cross-scenario nonanticipativity is imposed on them.

For fixed (w, z_ω), define the exact operational recourse value as

$$Q_\omega(w, z_\omega) = \min_{y_\omega} \left\{ C^{\mathrm{chg}}_\omega(y_\omega) : y_\omega \in \mathcal{Q}_\omega(w, z_\omega) \right\}. \tag{21}$$

The original scenario recourse value can then be decomposed as

$$\Phi_\omega(w) = \min_{z_\omega} \left\{ C^{\mathrm{bat}}_\omega(z_\omega) + Q_\omega(w, z_\omega) : z_\omega \in \mathcal{Z}_\omega \right\}, \tag{22}$$

where $\mathcal{Z}_\omega$ is the battery-choice feasible set defined by the one-option-per-route constraints in Section 3.

The lifted master is

$$\min_{w,z,\theta} \quad C^{\mathrm{fac}}(w) + \sum_{\omega\in\Omega} p_\omega \left[C^{\mathrm{bat}}_\omega(z_\omega) + \theta_\omega \right] \tag{23}$$

$$\text{s.t.} \quad z_\omega \in \mathcal{Z}_\omega, \qquad \omega \in \Omega, \tag{24}$$

$$\theta_\omega \geq \ell(w, z_\omega), \qquad \omega \in \Omega,\ \ell \in \mathcal{L}_\omega, \tag{25}$$

$$w_j \in \{0, 1\}, \qquad j \in \mathcal{J}, \tag{26}$$

$$z_{\omega,r,k} \in \{0, 1\}, \qquad \omega \in \Omega,\ r \in \mathcal{R},\ k \in \mathcal{K}. \tag{27}$$

The cuts in $\mathcal{L}_\omega$ are valid lower bounds on $Q_\omega(w, z_\omega)$. When the value function is represented exactly, the lifted master is an exact reformulation of the original two-stage model.

Proposition 1 (Exactness of the lifted reformulation)**.** *Suppose $\mathcal{L}_\omega$ represents $Q_\omega(w, z_\omega)$ exactly for each scenario ω. Then the lifted master formulation* (23)–(27) *is equivalent to the original two-stage formulation* (1)*.*

Proof. Fix any feasible solution of the original two-stage formulation. It consists of a facility layout w and, for each scenario ω, feasible battery and operating decisions (z_ω, y_ω). The same (w, z_ω) is feasible for the lifted master because z_ω is scenario dependent in both formulations. For each scenario, the optimal operating value associated

with (w, z_ω) is exactly $Q_\omega(w, z_\omega)$ by definition (21). Thus the lifted master can attain the same objective value as the original formulation.

Conversely, consider any feasible solution (w, z, θ) of the lifted master when the cuts represent the exact operational value functions. For each scenario, θ_ω is no smaller than $Q_\omega(w, z_\omega)$, and the definition of Q_ω implies the existence of an optimal operational decision y_ω whenever the value is finite. Combining w, z_ω, and these operational decisions gives a feasible solution of the original formulation with the same facility cost, battery cost, and operational value. Because z_ω is not forced to be the same across scenarios, the lifted master does not change the information structure of the original two-stage model. Therefore the two formulations have the same feasible objective values and the same optimum. □

The lifted reformulation isolates the remaining operational recourse and exposes the joint facility–battery resource structure used by the relaxation and cut-generation steps below.

4.2. Two-phase lower bounding and exact verification

After (w, z_ω) is fixed, the exact operational subproblem still contains two sources of difficulty. First, existing plug-in charger capacity constraints couple route-level charging decisions within a scenario. Second, plug-in charger occupancy remains binary. The two-phase method separates lower-bound generation from exact verification: Phase 1 relaxes shared capacity coupling and occupancy integrality to obtain route-level LP subproblems, and Phase 2 restores the exact scenario-level operational MIP.

4.2.1. Phase 1: route-level LP relaxation

Phase 1 relaxes the shared plug-in charger capacity constraints and the operational integrality requirements in $Q_\omega(w, z_\omega)$. Let $Q^R_\omega(w, z_\omega)$ be the resulting feasible set and $Q^R_\omega(w, z_\omega)$ its optimal value. Because the relaxed model removes constraints and integrality restrictions from a minimization subproblem, it provides a lower bound on the exact operational recourse. The removal of shared charger-capacity coupling also makes the relaxation additive across routes.

Proposition 2 (Phase 1 lower bound and route decomposition)**.** *For any fixed master solution* (w, z_ω)*,*

$$Q^R_\omega(w, z_\omega) \le Q_\omega(w, z_\omega). \tag{28}$$

Moreover, the Phase 1 relaxation decomposes exactly into independent route-level LP subproblems:

$$Q^R_\omega(w, z_\omega) = \sum_{r \in \mathcal{R}} Q^R_{\omega,r}(w, z_{\omega,r}). \tag{29}$$

Proof. For fixed (w, z_ω), every feasible solution of the exact operational subproblem is feasible for the Phase 1 relaxation because the relaxation removes shared plug-in charger capacity constraints and relaxes operational integrality. The Phase 1 feasible region therefore contains the exact feasible region. Since both problems minimize the same operating-cost expression over their respective feasible regions, the relaxed value cannot exceed the exact value, which proves (28).

After the shared plug-in charger capacity constraints are removed, the remaining charging-window, energy-balance, and wireless-availability constraints are route specific for fixed w and z_ω. The operating cost is also separable across the same route-level components. Thus the relaxed scenario problem is the Cartesian product of independent route-level LP feasible regions, and its objective is the sum of their route-level objectives. This gives the additive decomposition in (29). □

The route-level LPs produce dual information that can be aggregated into scenario-level lower-bounding cuts. A representative Phase 1 optimality cut has the form

$$\theta_\omega \ge \alpha_\omega + \sum_{j \in \mathcal{J}} \pi_{\omega j} w_j + \sum_{r \in \mathcal{R}} \sum_{k \in \mathcal{K}} \lambda_{\omega,r,k} z_{\omega,r,k}, \tag{30}$$

where the coefficients are obtained from the dual solutions of the route-level LP relaxations.

Proposition 3 (Validity of Phase 1 cuts for exact recourse)**.** *For a fixed scenario* ω*, any inequality that is a valid lower bound on* $Q^R_\omega(w, z_\omega)$ *over the lifted-master domain is also a valid lower bound on* $Q_\omega(w, z_\omega)$*. Moreover, if the Phase 1 relaxation is infeasible at a configuration* (w, z_ω)*, then the exact operational subproblem is also infeasible at that configuration.*

Proof. The first statement follows directly from Proposition 2: for every master-feasible (w, z_ω), a valid lower bound on $Q^R_\omega(w, z_\omega)$ is no greater than $Q^R_\omega(w, z_\omega)$, which is no greater than $Q_\omega(w, z_\omega)$. For the second statement, the exact operational feasible region is a subset of the Phase 1 relaxed feasible region. If the larger relaxed region is empty, then the exact feasible region must also be empty. □

Thus Phase 1 dual cuts can be used in the lifted master for exact Phase 2 verification, but Phase 1 objective values cannot be used as incumbent upper bounds. When a Phase 1 LP relaxation is infeasible, the corresponding Farkas information can be used to add a feasibility cut when available.

4.2.2. Phase 2: exact mixed-integer verification

Phase 2 restores the exact operational MIP for the current lifted-master solution. It evaluates $Q_\omega(w, z_\omega)$ directly, adds exact feasibility or optimality cuts when the master underestimates recourse, and updates the incumbent upper bound only when all scenario subproblems are exactly feasible. For a feasible lifted solution, the candidate upper bound is

$$UB = C^{\text{fac}}(w) + \sum_{\omega \in \Omega} p_\omega \left[C^{\text{bat}}_\omega(z_\omega) + Q_\omega(w, z_\omega) \right]. \tag{31}$$

This upper bound is not updated from Phase 1 relaxed values because those values may underestimate the exact operational cost.

The implementation also uses a Phase 2 relaxed-subproblem pass, referred to as RSP, before solving exact MIP subproblems. RSP solves LP relaxations of the Phase 2 scenario subproblems at the current master solution. Its infeasibility cuts and lower-bounding cuts can improve the Phase 2 master before exact MIP verification is attempted, using the same relaxation-to-exact validity logic as Proposition 3. RSP functions as a Phase 2 cut-generation enhancement; exact MIP verification remains the source of incumbent upper bounds.

4.3. Monotonicity-based strengthened cuts

The lifted reformulation exposes a joint resource order. Opening additional wireless facilities or selecting larger route batteries weakly preserves operational feasibility and cannot increase the minimum operating cost. The full method uses this order to strengthen feasibility and optimality cuts so that one recourse evaluation can cover a set of dominated configurations.

Index the battery options for each route in nondecreasing capacity order. For two configurations (w, z_ω) and $(\bar{w}, \bar{z}_\omega)$, write $(w, z_\omega) \preceq (\bar{w}, \bar{z}_\omega)$ if $w_j \le \bar{w}_j$ for all facilities j and the selected battery option for each route under z_ω is no larger than the selected option under $\bar{z}_\omega$.

Proposition 4 (Model-induced resource monotonicity)**.** *For a fixed service-plan scenario, suppose trip-level energy consumption is independent of the selected battery capacity. If $(w, z_\omega) \preceq (\bar{w}, \bar{z}_\omega)$, then every operating schedule feasible under (w, z_ω) can be transformed into a feasible operating schedule under $(\bar{w}, \bar{z}_\omega)$ with the same charging quantities. Consequently, $Q_\omega(\bar{w}, \bar{z}_\omega) \le Q_\omega(w, z_\omega)$.*

Proof. Take any feasible operating solution under (w, z_ω). Keep all plug-in, SWC, DWC, and overnight charging quantities unchanged. Opening additional wireless facilities does not invalidate any wireless-availability constraint, because the original charging amounts remain within the old upper bounds and the new facility variables weakly increase these bounds.

It remains to check battery upgrades, because the absolute SoC interval changes from $[\underline{\rho} B_{\omega,r}, \overline{\rho} B_{\omega,r}]$ to $[\underline{\rho} \bar{B}_{\omega,r}, \overline{\rho} \bar{B}_{\omega,r}]$. For each route r, let $\Delta_r = \bar{B}_{\omega,r} - B_{\omega,r} \ge 0$ and increase every arrival and departure energy state on that route by $\overline{\rho}\Delta_r$. All station and link energy-balance equations remain valid because the same shift is applied to both sides of each balance equation, and trip-level energy consumptions d_a, d^0_m, d^N_m are unchanged by assumption. The upper SoC bounds hold by construction. The lower SoC bounds also hold since

$$e + \overline{\rho}\Delta_r \ge \underline{\rho} B_{\omega,r} + \overline{\rho}\Delta_r \ge \underline{\rho} \bar{B}_{\omega,r}.$$

The initial depot boundary equation shifts from $\overline{\rho} B_{\omega,r} - d^0_m$ to $\overline{\rho} \bar{B}_{\omega,r} - d^0_m$, the return-to-depot lower bound is preserved by the same lower-bound argument, and the overnight charging balance remains valid because both $\overline{\rho} B_{\omega,r}$ and the terminal departure energy increase by $\overline{\rho}\Delta_r$. Thus the transformed operating solution is feasible under the enlarged configuration and has the same charging cost. Minimizing over operating schedules gives $Q_\omega(\bar{w}, \bar{z}_\omega) \le Q_\omega(w, z_\omega)$. □

Given an evaluated configuration (w^*, z^*_ω), let $k^*_{\omega,r}$ be the selected battery option on route r and define

$$I_\omega(w, z_\omega; w^*, z^*_\omega) = \sum_{j: w^*_j = 0} w_j + \sum_{r \in \mathcal{R}} \sum_{k \in \mathcal{K}: k > k^*_{\omega,r}} z_{\omega,r,k}. \tag{32}$$

The indicator is zero exactly for configurations that do not add a facility and do not upgrade any route battery relative to (w^*, z^*_ω), that is, configurations dominated by the evaluated one.

Proposition 5 (Strengthened feasibility cut)**.** *Suppose the exact operational subproblem for scenario ω is infeasible at (w^*, z^*_ω). Then*

$$I_\omega(w, z_\omega; w^*, z^*_\omega) \geq 1 \tag{33}$$

is a valid feasibility cut for scenario ω.

Proof. Consider any configuration with $I_\omega(w, z_\omega; w^*, z^*_\omega) = 0$. By construction, it opens no facility that is closed in w^* and selects no route battery option larger than the option selected in z^*_ω. Therefore $(w, z_\omega) \preceq (w^*, z^*_\omega)$. If such a dominated configuration were operationally feasible, then Proposition 4 would imply that the enlarged configuration (w^*, z^*_ω) is also feasible. This contradicts the assumed infeasibility of (w^*, z^*_ω). Thus every feasible master configuration must satisfy (33). □

Proposition 6 (Strengthened optimality cut)**.** *Suppose the exact operational subproblem for scenario ω is feasible at (w^*, z^*_ω) with value Q^*_ω. Let L_ω be any valid global lower bound satisfying $L_\omega \leq Q_\omega(w, z_\omega)$ for every master-feasible configuration (w, z_ω). Define $\delta_\omega = Q^*_\omega - L_\omega \geq 0$. Then*

$$\theta_\omega \geq Q^*_\omega - \delta_\omega I_\omega(w, z_\omega; w^*, z^*_\omega) \tag{34}$$

is a valid optimality cut for scenario ω.

Proof. If $I_\omega(w, z_\omega; w^*, z^*_\omega) = 0$, then $(w, z_\omega) \preceq (w^*, z^*_\omega)$. Proposition 4 implies that the evaluated configuration has at least as much resource as (w, z_ω), so its exact operating cost is no larger:

$$Q_\omega(w, z_\omega) \geq Q_\omega(w^*, z^*_\omega) = Q^*_\omega.$$

The right-hand side of (34) is then Q^*_ω, which is a valid lower bound on $Q_\omega(w, z_\omega)$.

If $I_\omega(w, z_\omega; w^*, z^*_\omega) \geq 1$, then

$$Q^*_\omega - \delta_\omega I_\omega(w, z_\omega; w^*, z^*_\omega) \leq Q^*_\omega - \delta_\omega = L_\omega.$$

Because L_ω is a valid global lower bound, the right-hand side of (34) is again no greater than $Q_\omega(w, z_\omega)$. Thus the inequality is valid for all master-feasible configurations. □

The feasibility cut and optimality cut are direction selective: they strengthen the master only in the part of the configuration space justified by the joint facility–battery order. Within the lifted two-phase decomposition, the facility–battery partial order lets one exact recourse evaluation eliminate or bound multiple dominated configurations.

4.4. Complete solution procedure

Algorithm 1 summarizes the full method. Phase 1 initializes and strengthens the lifted master using route-level LP lower-bounding separation. Phase 2 then reuses the lifted master, optionally performs RSP separation, solves exact scenario MIPs, and adds exact feasibility and optimality cuts. Only exact Phase 2 feasible evaluations can update the incumbent upper bound.

The implementation stops when the prescribed optimality tolerance or time limit is reached. The finite-convergence statement below is separate from such implementation tolerances and assumes exact solves.

Theorem 1 (Finite convergence under exact separation)**.** *Assume that the lifted master is solved exactly at every iteration, exact Phase 2 scenario subproblems are solved exactly, and all feasibility and optimality cuts added in Phase 2 are valid. Then the exact Phase 2 separation loop terminates finitely with an optimal solution of the lifted reformulation and hence of the original two-stage formulation.*

Proof. The lifted master contains finitely many binary facility–battery configurations because $\mathcal{J}$, Ω, $\mathcal{R}$, and $\mathcal{K}$ are finite. Consider an iteration of the exact Phase 2 loop. If the evaluated configuration is infeasible for some scenario, a valid feasibility cut excludes that infeasible configuration and possibly additional dominated infeasible configurations. If the evaluated configuration is feasible but its exact recourse value is underestimated by the master, a valid optimality cut raises the master approximation for that configuration, so the same unresolved underestimation cannot recur unchanged.

Since there are finitely many binary configurations, the loop cannot generate an infinite sequence of distinct unresolved configurations. When no violated exact feasibility or optimality cut exists, the lifted master objective equals the objective value obtained by exact recourse evaluation for the incumbent configuration. At that point the master lower bound and the exact incumbent upper bound coincide, so the incumbent is optimal for the lifted reformulation. Proposition 1 then implies optimality for the original two-stage formulation. □

Input : Scenario data, candidate wireless facilities, candidate battery options, tolerance ε, time limit T
Output: Facility layout w, scenario battery configurations z, incumbent upper bound, and final lower bound

Initialize the lifted master with battery-selection constraints and valid recourse lower bounds;

`// Phase 1: route-level LP lower-bounding separation`
repeat
  Solve the lifted master;
  Solve route-level LP relaxations for each scenario at the current (w, z_ω);
  Add LP-dual lower-bounding cuts and available relaxation-feasibility cuts;
until *no violated Phase 1 cut is found or the Phase 1 stopping rule is met*;

`// Phase 2: exact mixed-integer verification`
repeat
  Solve the lifted master;
  Optionally solve RSP LP relaxations and add valid Phase 2 RSP cuts;
  Solve exact scenario operational MIP subproblems at the current (w, z_ω);
  Add exact feasibility cuts, including strengthened feasibility cuts when applicable;
  Add exact monotonicity-based optimality cuts when recourse is underestimated;
  Update the incumbent upper bound only if all exact scenario MIPs are feasible;
until *the incumbent gap is at most ε or the time limit is reached*;
return *Best incumbent and final lower bound*;

Algorithm 1: Full reformulation-enabled integer L-shaped method

5. Computational Results

This section evaluates the proposed reformulation and solution method on Tower-derived wireless charging planning instances. The experiments use the bus network as a realistic source of shared stops, shared links, and route-level service plans, while the main purpose of the section is to validate algorithmic performance.

5.1. Experimental design and instance families

The test instances are derived from the Tower Transit bus network in Singapore's Jurong area. The resulting data set provides a realistic spatial structure with shared stops, shared road links, and multiple routes that may use the same wireless charging infrastructure. We use these instances as a computational test bed rather than as a public benchmark.

Each scenario represents an alternative medium-term service plan for the same physical network, not random same-day scheduling noise. Scenario-dependent data include the operated routes, bus duties, timetables, energy requirements, and charging opportunities induced by the corresponding service plan. The first-stage wireless charging facilities are shared across all scenarios, whereas battery and charging-operation decisions adapt after the service plan is realized.

We use a compact instance code of the form `N-k`, where `N` indicates the network scale and k indicates the number of service-plan scenarios. Specifically, `S`, `M`, `B`, and `L` denote the small, medium, base, and large Tower-derived networks, respectively. The capacity-coupling experiment appends `Cq` to the base 10-scenario instance, where q indexes the number of existing plug-in fast chargers. The network-scale family fixes the scenario count at three and varies the physical network scale. The scenario-count family fixes the base network and varies the number of service-plan scenarios. Motivated by the relaxation-verification structure in Section 4.2, the capacity-coupling family varies plug-in fast-charger capacity on `B-10` to change how strongly residual fast-charging demand couples route-level recourse decisions. Table 1 summarizes the resulting instance families and their experimental roles.

Travel times are computed from link distances using a nominal operating speed of 30 km/h and then rounded to the one-minute scheduling grid.[1] The remaining parameters specify battery costs, charging rates, charging efficiencies, electricity prices, facility costs, and state-of-charge limits. Table 2 reports the main parameter values used in the computational experiments.

The benchmark ladder consists of five methods: the direct extensive-form MILP, baseline integer L-shaped decomposition, lifted integer L-shaped decomposition, lifted two-phase method, and the full method. The direct

[1]The one-minute lower bound avoids zero-duration movements on short links and absorbs acceleration, deceleration, signal delay, and station-approach time.

extensive-form MILP optimizes the deterministic equivalent with a commercial solver. The baseline integer L-shaped decomposition applies the classical scenario decomposition induced by the original two-stage model. The lifted integer L-shaped decomposition uses the lifted master representation without the two-phase relaxation. Lifted two-phase adds the Phase 1 lower-bounding and Phase 2 verification structure. The full method further adds the monotonicity-based strengthened cuts.

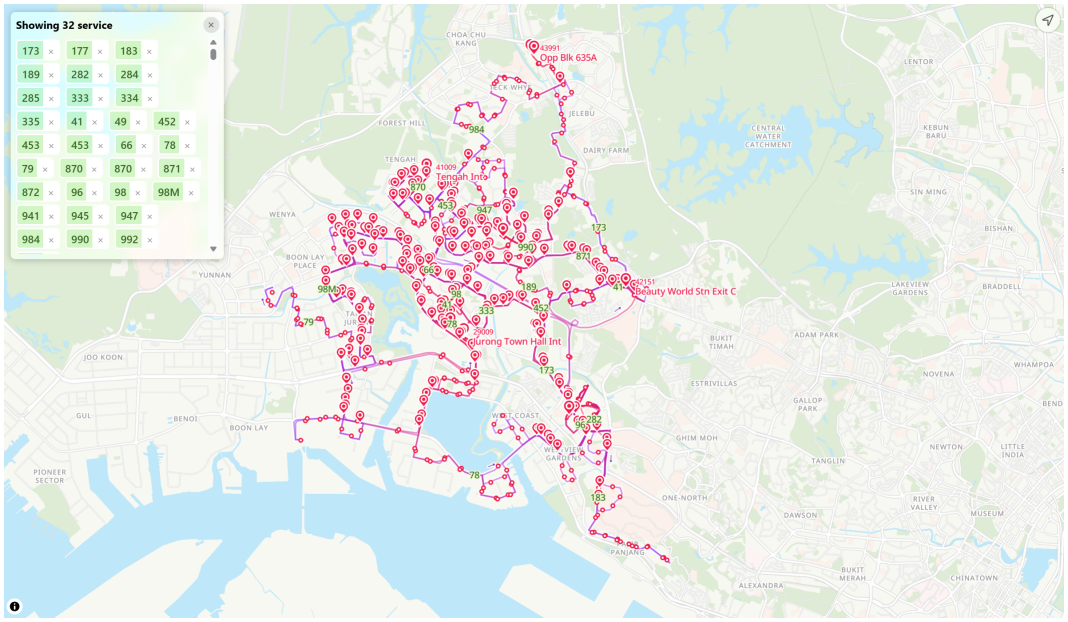


Figure 1: Spatial layout of the base Tower-derived network in the Jurong area.

Table 1: Main instance families and experimental use

Instance family	Data source or variant	Varied dimension	Primary use	Compared methods
`S-3`, `M-3`, `B-3`, `L-3`	`tower_small`, `tower_medium`, `tower`, `large`	Network scale, with three scenarios fixed	Network-scale ablation	Direct MILP, baseline integer L-shaped, lifted integer L-shaped, lifted two-phase, full method
`B-k`, $k \in \{3, 5, 10, 20, 40\}$	`tower`, `tower_5`, `tower_10`, `tower_20`, `tower_40`	Scenario count, with base network fixed	Scenario-count scalability	Direct MILP, lifted two-phase, full method
`B-10-Cq`, $q \in \{1, 2, 3, 4, 5\}$	Capacity variants of `tower_10`	Existing plug-in fast-charger units, with base network and 10 scenarios fixed	Relaxation-gap stress test	Lifted two-phase, full method

`B-3` is the representative base instance used for solution interpretation; larger q indicates more existing plug-in fast-charger units.

5.2. Solution characteristics on the base Tower instance

This subsection will use `B-3` as the representative planning solution.

The solution-summary table will report selected SWC facilities, selected DWC links, cost components, objective value, and battery-capacity summaries.

Representative selected stops or links will show that the model places wireless infrastructure on shared parts of the bus network.

The planning interpretation is that the model recommends a network-level wireless charging layout under service-plan uncertainty rather than a single-day charging schedule.

5.3. Network-scale algorithmic ablation

The network-scale ablation compares five methods on `S-3`, `M-3`, `B-3`, and `L-3`.

The largest reported instances are intended to show where direct optimization and simpler decomposition variants stop being computationally effective.

The mechanism contrast is that lifted two-phase still generates many weak verification cycles on `B-3`, whereas the strengthened full method converges quickly with only a few Phase 2 iterations.

The phase-level evidence supports the interpretation that the full method controls both first-stage bound improvement and second-stage mixed-integer verification.

5.4. Scenario-count scalability

This subsection uses the `B-k` family, with $k \in \{3, 5, 10, 20, 40\}$, to test scenario-count scalability on the base Tower-derived network. The comparison is limited to the direct extensive-form MILP, lifted two-phase method, and full method because the simpler integer L-shaped variants already fail in the network-scale ablation.

The main diagnostic is how total time, Phase 1 time, Phase 2 time, and iteration counts change as more scenario-specific recourse problems are added to the same first-stage infrastructure plan.

Table 2: Parameter settings for the computational experiments

Parameter	Description	Value	Source
Battery and state of charge			
c^{bat}	Daily-equivalent battery capacity cost	0.148 USD/kWh/day	BloombergNEF price amortized over Wang et al. battery life (BloombergNEF, 2025; Wang et al., 2024)
$\mathcal{K}_r$	Candidate route-level battery options	10-kWh increments within instance bounds	Case-data preprocessing
$\underline{\rho}, \overline{\rho}$	Minimum and maximum state-of-charge fractions	0.2, 0.8	Wireless bus planning literature (Li et al., 2024)
Charging operation			
r^{F}	Plug-in fast-charging rate	2.5 kWh/min	Najafi et al. (2025)
r^{SWC}	Stationary wireless charging rate	1.0 kWh/min	Wang et al. (2024)
r^{DWC}	Dynamic wireless charging rate	1.0 kWh/min	Wang et al. (2024)
η^{F}	Plug-in fast-charging efficiency	0.90	Li et al. (2024)
η^{SWC}	Stationary wireless charging efficiency	0.85	Sun et al. (2024)
η^{DWC}	Dynamic wireless charging efficiency	0.80	Sun et al. (2024)
a_s	Existing plug-in charger units at each fast-charging location	4 chargers	Case-study assumption
Infrastructure and electricity cost			
T^{fac}	Facility amortization horizon	30 years	Wang et al. (2024)
c_j^{fac}, $j \in \mathcal{J}^{\text{SWC}}$	Daily-equivalent cost of an SWC facility	2.16 USD/day	Wang et al. (2024)
c_j^{fac}, $j \in \mathcal{J}^{\text{DWC}}$	Daily-equivalent cost of a DWC link of length d_j km	$9d_j + 0.90$ USD/day	Wang et al. (2024)
c^{day}	Daytime electricity price	0.1922 USD/kWh	Singapore tariff data (Singapore Department of Statistics, 2026)
c^{night}	Nighttime electricity price	0.1233 USD/kWh	Singapore tariff data (Singapore Department of Statistics, 2026)

Table 3: Network-scale ablation under three service-plan scenarios

Instance	Direct MILP	Baseline integer L-shaped	Lifted integer L-shaped	Lifted two-phase	Full method
S-3	Conv., 4.0s	TL, 98.3%	TL, 80.0%	Conv., 0.3s	Conv., 0.4s
M-3	TL, 0.28%	skipped	skipped	Conv., 26.2s	Conv., 21.6s
B-3	OOM, 0.27%	skipped	skipped	TL, 0.60%	Conv., 45.4s
L-3	TODO	skipped	skipped	TODO	TODO

Conv. = converged within the 0.01% tolerance; TL = 7200s time limit with incumbent;
OOM = Gurobi memory error 10001. Reported percentages are final relative gaps. Prior-run values are placeholders to be rerun.

Table 4: Mechanism contrast on B-3

Method	Status	Gap (%)	Time (s)	Iterations
Lifted two-phase	TL	0.595	7211.8	548
Full method	Conv.	0.010	45.4	9/3

Full-method iterations are reported as Phase 1/Phase 2 iterations.

Table 5: Scenario-count scalability on the base Tower-derived network

Instance	Direct MILP	Lifted two-phase	Full method	Phase 1/2 (s)	Iters P1/P2
B-3	OOM, 0.27%	TL, 0.60%	Conv., 45.4s	29.7/15.7	9/3
B-5	TODO	TODO	Conv., 52.8s	36.3/16.5	6/3
B-10	TODO	TODO	Conv., 126.6s	68.1/58.5	3/4
B-20	TODO	TODO	Conv., 356.3s	145.6/210.7	9/4
B-40	TODO	TODO	Conv., 1702.1s	447.3/1254.8	12/2

All entries are placeholders from prior runs and should be rerun under the final settings.

5.5. Capacity-coupling and relaxation-gap experiment

This subsection uses B-10-C1 through B-10-C5 to test the effect of shared plug-in fast-charger capacity on the relaxation-verification structure. Within this subsection, we denote these five variants by C1–C5; all five are built from B-10, and only the number of existing plug-in fast chargers is varied. This changes how residual fast-charging demand, after accounting for wireless charging opportunities and battery choices, couples the route-level recourse problems. The experiment is designed around the Phase 1 relaxation in Proposition 2, which removes shared plug-in fast-charger capacity coupling to obtain route-level LP subproblems. When capacity is tight, the Phase 1 relaxation is expected to be less aligned with Phase 2; when capacity is loose, the two feasible regions should be closer. The comparison is restricted to lifted two-phase and the full method because both use the same lifted reformulation, Phase 1 relaxation, and exact Phase 2 verification; the main difference is the monotonicity strengthening. The main computational question is how the relaxation gap and Phase 2 verification burden change with capacity, and whether monotonicity strengthening remains effective as capacity coupling becomes tighter.

Table 6: Capacity-coupling and relaxation-gap stress test on B-10

Instance	Lifted two-phase	Full method	Initial P2 gap	P1 time	P2 master time	Exact subproblem time	P1/P2 cuts
C1	TODO	TODO	TODO	TODO	TODO	TODO	TODO
C2	TODO	TODO	TODO	TODO	TODO	TODO	TODO
C3	TODO	TODO	TODO	TODO	TODO	TODO	TODO
C4	TODO	TODO	TODO	TODO	TODO	TODO	TODO
C5	TODO	TODO	TODO	TODO	TODO	TODO	TODO

All entries are placeholders for the final capacity-coupling experiment.

6. Conclusion

This paper studies a two-stage wireless charging facility-location problem for electric-bus systems under service-plan uncertainty. The model treats SWC and DWC deployment as strategic here-and-now decisions, while route-level battery selection and representative-day charging operations adapt after each service-plan scenario is realized. The resulting recourse structure captures how wireless charging opportunities affect battery requirements and residual demand for limited shared plug-in fast chargers.

The main methodological contribution is an exact decomposition framework for this shared-resource mixed-integer recourse structure. The method uses an exact lifted reformulation that keeps scenario-dependent battery choices adaptive, a nested scenario–route relaxation–verification decomposition, and joint-resource direction-selective monotonicity cuts. Phase 1 route-level LP relaxations obtain valid lower bounds and cuts, Phase 2 restores exact mixed-integer operational verification, and the strengthened full method applies monotonicity cuts where they are valid for exact recourse. Under exact master and recourse separation, the complete procedure converges to an optimal solution of the original two-stage formulation.

The computational study is designed to show how each algorithmic component contributes to scalability. The base planning instance illustrates the type of wireless facility layout and adaptive battery decisions produced by the model. The ablation, scenario-scaling, and capacity-coupling experiments then evaluate when the direct extensive-form MILP and simpler decomposition variants become insufficient and when the full method remains effective.

Several limitations define natural extensions. The current experiments use constructed service-plan scenarios, treat plug-in fast-charging infrastructure as exogenous, and do not endogenize the feedback from battery weight to route-level energy consumption. Future work can develop richer scenario-generation models, jointly optimize

wireless and plug-in charging expansion, incorporate operational uncertainty, and apply the lifted two-phase decomposition idea to other stochastic infrastructure-planning problems with shared-resource recourse.

Appendix A. Model Notation

Table A.7: Summary of notation

Symbol	Description
Sets and indices	
Ω	Set of service-plan scenarios, indexed by ω
$\mathcal{S}$	Set of stations, indexed by s
$\mathcal{L}$	Set of directed links between consecutive stations, indexed by l
C	Set of existing interchanges or hubs equipped with plug-in chargers, $C \subseteq \mathcal{S}$
$\mathcal{R}$	Set of routes, indexed by r
$\mathcal{K}$	Set of candidate battery-capacity options, indexed by k
$\mathcal{M}_{\omega,r}$	Set of bus trip chains on route r under scenario ω, indexed by m
$\mathcal{V}_{\omega,r}$	Set of station visits on route r under scenario ω, indexed by v
$\mathcal{V}^{\mathrm{P}}_{\omega,r}$	Set of plug-in charging visits on route r under scenario ω
$\mathcal{V}^{\mathrm{S}}_{\omega,r}$	Set of SWC-eligible station visits on route r under scenario ω
$\mathcal{A}_{\omega,r}$	Set of directed link traversals on route r under scenario ω, indexed by a
$\mathcal{T}_{\omega}$	Relevant plug-in charging time periods under scenario ω
$\mathcal{T}_{v}$	Time periods during which plug-in charging visit v can occupy a plug-in charger
Parameters	
p_{ω}	Probability of scenario ω
c^{S}_{s}	Representative-day equivalent cost of installing SWC at station s
c^{D}_{l}	Representative-day equivalent cost of installing DWC on link l
c^{bat}	Representative-day equivalent battery capacity cost per kWh
$c^{\mathrm{day}}, c^{\mathrm{night}}$	Daytime and overnight electricity prices
$n_{\omega,r}$	Fleet-size requirement for route r under scenario ω
β_k	Battery capacity value represented by option k
d_a	Energy consumption associated with directed link traversal a
d^{0}_{m}	Boundary energy consumption from the depot to the first station visit of trip chain m
d^{N}_{m}	Boundary energy consumption from the terminal station visit of trip chain m back to the depot
U_s	Number of existing plug-in chargers at station s
κ	Grid-side energy delivered by one occupied plug-in time slot
$\bar{q}^{\mathrm{S}}_{v}$	Maximum SWC energy available during station visit v
$\bar{q}^{\mathrm{D}}_{a}$	Maximum DWC energy available during directed link traversal a
$\eta^{\mathrm{P}}, \eta^{\mathrm{S}}, \eta^{\mathrm{D}}$	Charging efficiencies for plug-in charging, SWC, and DWC
$\underline{\rho}, \bar{\rho}$	Minimum and maximum state-of-charge fractions
Derived quantities and functions	
$s(v)$	Physical station associated with station visit v
$\ell(a)$	Directed link associated with traversal a
$\mathrm{tail}(a)$	Departure station visit of traversal a
$\mathrm{head}(a)$	Arrival station visit of traversal a
v^{0}_{m}	First station visit of trip chain m
v^{N}_{m}	Terminal station visit of trip chain m
Decision variables	
w^{S}_{s}	1 if SWC is installed at station $s \in \mathcal{S} \setminus C$, 0 otherwise
w^{D}_{l}	1 if DWC is installed on link $l \in \mathcal{L}$, 0 otherwise
$z_{\omega,r,k}$	1 if route r uses battery capacity option k in scenario ω, 0 otherwise
$B_{\omega,r}$	Selected battery capacity for route r in scenario ω
y_{ω}	Vector of scenario-level charging, plug-in occupancy, and energy-state decisions in scenario ω
q^{P}_{v}	Plug-in charging energy assigned during plug-in visit v
q^{S}_{v}	SWC energy assigned during SWC-eligible station visit v
q^{D}_{a}	DWC energy assigned to directed link traversal a
q^{N}_{m}	Overnight charging energy assigned after trip chain m returns to the depot
$e^{\mathrm{arr}}_{v}, e^{\mathrm{dep}}_{v}$	Battery energy state at arrival and departure of station visit v
$x_{v\tau}$	1 if plug-in charging visit v occupies one plug-in charger at time τ, 0 otherwise

Appendix B. Proofs

Appendix C. Additional Computational Results

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